

M6_Assignment

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Q2 Netflix Subscribers

Q2.1 Loading Packages and Data

Load the fpp3 and tidyverse packages. Create an object that captures the data called Netflix.

```
library(fpp3)
```

```
Warning: package 'fpp3' was built under R version 4.4.2
```

```
Registered S3 method overwritten by 'tsibble':
```

```
  method          from  
as_tibble.grouped_df dplyr
```

```
-- Attaching packages ----- fpp3 1.0.1 --
```

```
v tibble      3.2.1    v tsibble      1.1.5  
v dplyr       1.1.4    v tsibbledata 0.4.1  
v tidyr       1.3.1    v feasts      0.4.1  
v lubridate   1.9.3    v fable       0.4.1  
v ggplot2     3.5.1
```

```
Warning: package 'tsibble' was built under R version 4.4.2
```

```
Warning: package 'tsibbledata' was built under R version 4.4.2
```

```
Warning: package 'feasts' was built under R version 4.4.2
```

Warning: package 'fabletools' was built under R version 4.4.2

Warning: package 'fable' was built under R version 4.4.2

```
-- Conflicts ----- fpp3_conflicts --
x lubridate::date()      masks base::date()
x dplyr::filter()        masks stats::filter()
x tsibble::intersect()   masks base::intersect()
x tsibble::interval()    masks lubridate::interval()
x dplyr::lag()            masks stats::lag()
x tsibble::setdiff()     masks base::setdiff()
x tsibble::union()       masks base::union()
```

```
library(tidyverse)
```

```
-- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
v forcats 1.0.0      v readr    2.1.5
v purrr   1.0.2      v stringr 1.5.1
```

```
-- Conflicts ----- tidyverse_conflicts() --
x dplyr::filter()      masks stats::filter()
x tsibble::interval()  masks lubridate::interval()
x dplyr::lag()          masks stats::lag()
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become
```

```
Netflix <- read_csv("https://jagelves.github.io/Data/Netflix.csv")
```

Rows: 46 Columns: 2

```
-- Column specification -----
Delimiter: ","
chr (1): Quarter
dbl (1): Subscribers
```

i Use `spec()` to retrieve the full column specification for this data.

i Specify the column types or set `show\_col\_types = FALSE` to quiet this message.

Q2.2 Creating a tsibble

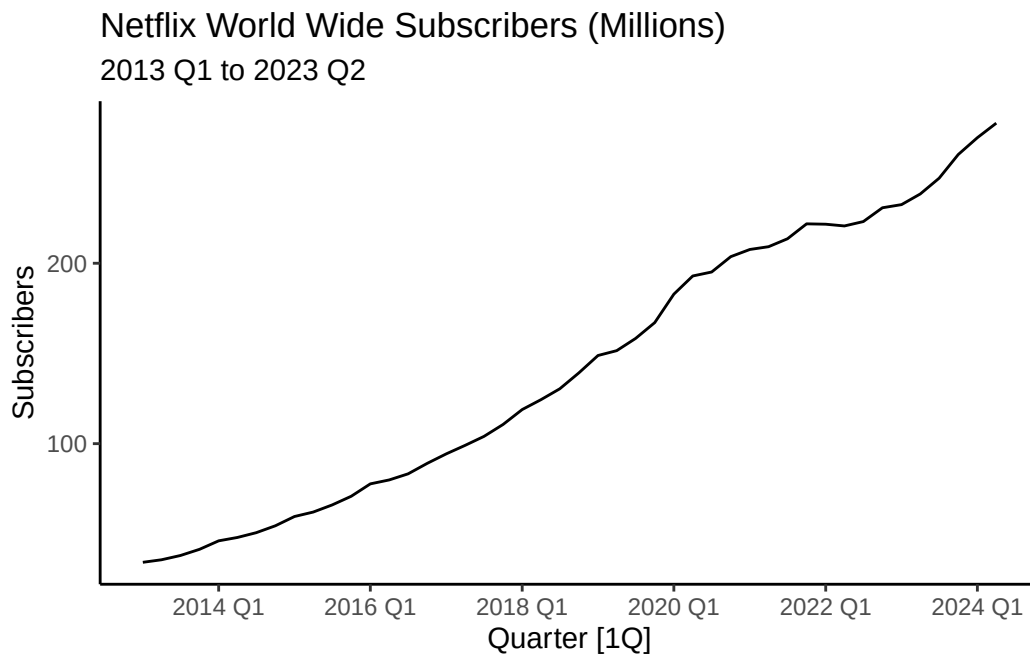
Use piping to: - coerce the Quarter variable to a date that captures the year and quarter. - coerce the data to a tsibble Name the resulting tsibble Netflix_ts.

```
Netflix %>%  
  mutate(Quarter=yearquarter(Quarter)) %>%  
  as_tsibble(index=Quarter) -> Netflix_ts
```

Q2.3 Plot

Use the autoplot() function to plot the Subscriber time series. Add a title that reads “Netflix World Wide Subscribers (Millions)”, and a subtitle that reads “2013 Q1 to 2023 Q2”.

```
Netflix_ts %>% autoplot(Subscribers) + theme_classic() +  
  labs(title= "Netflix World Wide Subscribers (Millions)",  
        subtitle = "2013 Q1 to 2023 Q2")
```



Q2.4 Test and Train Sets

Use the filter_index() function to create train and test sets. Name the sets train_net and test_net, respectively. The train set should go from the first quarter of 2013 to the fourth

quarter of 2021. The test set should go from the first quarter of 2022 to the second quarter of 2023.

```
train_net <- filter_index(.data=Netflix_ts,"2013 Q1"~"2021 Q4")
test_net <- filter_index(.data=Netflix_ts,"2022 Q1"~"2023 Q2")
```

Q2.5 CV

Perform cross-validation using the train set. Use the `stretch_tsibble()` function with the `.init` argument set to 12 and the `.step` argument set to four. The following models should be estimated:

- ETS with additive error, damped additive trend, and additive seasonality.
- ETS with no arguments. Algorithm.
- TSLM with trend and trend squared.
- TSLM with trend, trend squared, and seasonality. Forecast four periods and test accuracy against the test set.

```
library(gt)
```

Warning: package 'gt' was built under R version 4.4.2

```
train_net %>% stretch_tsibble(.init = 12, .step=4) %>%
  model(
    DAMPEDS=ETS(Subscribers ~ error("A") + trend("Ad") + season("A")),
    ALGO=ETS(Subscribers),
    LS = TSLM(Subscribers ~ trend()+I(trend()^2)),
    LSS = TSLM(Subscribers ~ trend()+I(trend()^2)+season()))%>%
  forecast(h = 4) %>%
  accuracy(test_net)
```

Warning: The future dataset is incomplete, incomplete out-of-sample data will be treated as missing. 24 observations are missing between 2016 Q1 and 2021 Q4

```
# A tibble: 4 x 10
  .model .type    ME  RMSE  MAE    MPE  MAPE  MASE  RMSSE  ACF1
  <chr>  <chr> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl>
1 ALGO   Test   -13.3  13.7  13.3  -5.93  5.93   NaN   NaN  0.148
2 DAMPEDS Test   -12.2  12.5  12.2  -5.45  5.45   NaN   NaN  0.0693
3 LS     Test   -29.3  30.2  29.3 -13.1  13.1   NaN   NaN  0.222
4 LSS    Test   -29.5  30.1  29.5 -13.2  13.2   NaN   NaN  0.221
```

Q2.6 Best Model

Refer to the table constructed in Q2.5. What is the best model according to the RMSE?

```
# DAMPEDS - ETS(A,Ad,A)
```

Q2.7 Fit Object

Use the entire data (Netflix_ts) to create an object called fit that estimates only the top ETS model and the top TSLM by RMSE. Refer to the table in Q2.5. Include your code below:

```
fit <- Netflix_ts %>%  
  model(DAMPEDS=ETS(Subscribers ~ error("A") + trend("Ad") + season("A")),  
        LSS = TSLM(Subscribers ~ trend()+I(trend()^2)+season()))
```

Q2.8 Information Criterion

Use the fit object to obtain the AIC, AICc, and BIC. You can start with the fit object and pipe to the glance() function to do this. Pipe one more time to select only .model, AIC, AICc, and BIC.

```
fit %>%  
  glance() %>%  
  select(".model", "AIC", "AICc", "BIC")
```

```
# A tibble: 2 x 4  
  .model    AIC  AICc   BIC  
  <chr>    <dbl> <dbl> <dbl>  
1 DAMPEDS  290.  296.  308.  
2 LSS      202.  204.  214.
```

Q2.9 Forecast

Use the fit object to forecast four quarters into the future (h=4). Upload a screenshot of the forecast output (table not plot) that includes the model used, the quarter, the distribution of the forecast, and the mean.

```
fit %>%  
  forecast(h=4)
```

```
# A fable: 8 x 4 [1Q]
# Key:      .model [2]
#           .model Quarter
#           <chr>      <qtr>
1 DAMPEDS 2024 Q3
2 DAMPEDS 2024 Q4
3 DAMPEDS 2025 Q1
4 DAMPEDS 2025 Q2
5 LSS      2024 Q3
6 LSS      2024 Q4
7 LSS      2025 Q1
8 LSS      2025 Q2
# i 2 more variables: Subscribers <dist>, .mean <dbl>
```

Q2.10 Plot2

Upload a screenshot of Netflix subscribers and the ETS and LSTM forecasts below. The title should be “Netflix Subscribers”, the subtitle “2013 Q1 to 2024 Q2 (estimated)”, include 90% prediction intervals, and omit the x and y labels.

```
fit %>%
  forecast(h = 4) %>%
  autoplot(Netflix_ts, level=90)+
  labs(x="", y="",
       title = "Netflix Subscribers",
       subtitle = "2013 Q1 to 2024 Q2 (estimated)") + theme_classic()
```

Netflix Subscribers
2013 Q1 to 2024 Q2 (estimated)

